22CH203 - ENGINEERING CHEMISTRY II

UNIT III – METAL CORROSION AND ITS PREVENTION

Lesson Plan 1.1.
Topic: Origin of Corrosion

References: 1. Chemistry for Engineering Students (Lawrence S. Brown, Thomas A. Holme), 4th Edition- Chapter 13.
2. Inorganic Chemistry (Gary L.Miessler Donald A.Tarr), 3rd Edition- Chapter 17.

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1. INTRODUCTION:

Corrosion is a natural process that leads to the deterioration of materials, particularly metals, through chemical reactions with their environment. One of the most common forms of corrosion is rusting, where iron reacts with moisture and oxygen to form iron oxide, weakening the metal. We encounter the effects of corrosion daily—whether it's the rust on bicycles, the tarnishing of silver, or the corrosion of water pipes (as shown in Figure 1). The term corrosion is sometimes also applied to degradation of plastics, concrete, wood and materials.

Over time, corrosion can cause significant damage to materials like cars, bridges, and household appliances. Beyond personal inconvenience, corrosion has a broader economic impact. It is estimated that corrosion costs nations billions annually in repairs, replacements, and lost productivity. This burden on industries such as transportation, infrastructure, and manufacturing directly affects the GDP of nations. The costs of maintaining and replacing corroded materials are often passed on to consumers, further straining the economy.

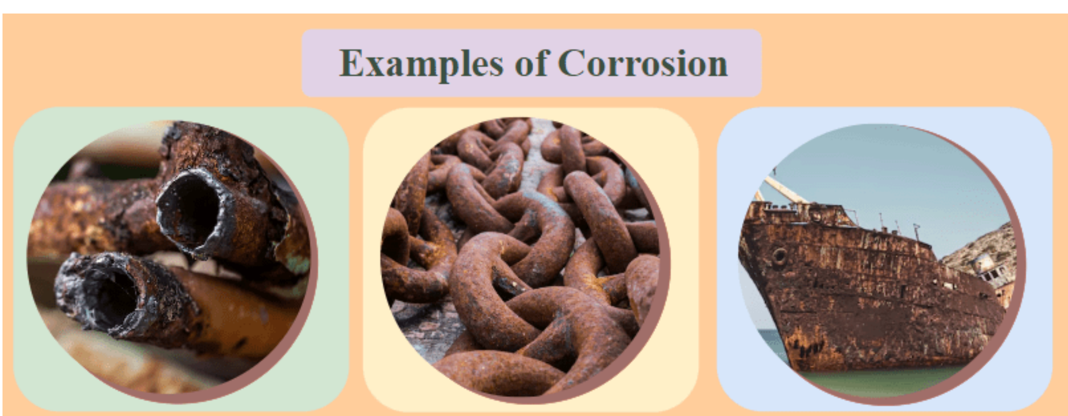

Figure 1. Examples of corrosion.

1.1. Definition:

"Corrosion is defined as the gradual destruction or deterioration of metals or alloys by the chemical or electrochemical reaction with its environment."

Corrosion is the gradual deterioration of metals or alloys through chemical or electrochemical reactions with the environment. While oxidation refers specifically to the loss of electrons in a substance, corrosion is a broader process involving oxidation that leads to material breakdown. Reactions encompass any chemical change, including oxidation and corrosion.

Corrosion is not instant; it occurs over time due to the gradual nature of material degradation. Various environmental factors, such as moisture, temperature, oxygen, pollutants, and saltwater, accelerate corrosion by facilitating the chemical and electrochemical processes that lead to material weakening.

1.2. Origin of Corrosion

1.2.1. Causes of Corrosion:

Metals occur in nature in two different forms.

- Native state.
- Combined state.

1.2.2. Native State:

The metals occurring in native (or) free (or) uncombined states are non-reactive with the environment. They are noble metals that exist as such in the earth's crust. They have very good corrosion resistance.

Examples: Gold (Au), Platinum (Pt), Silver (Ag).

Noble metals, by virtue of their stable electron configuration, strong metallic bonding, inertness, and low reactivity, are highly resistant to corrosion and can exist in their native state in the Earth's crust without undergoing significant chemical reactions. This makes them valuable for various applications, including jewelry, electronics, and catalytic processes.

1.2.3. Combined State:

Except noble metals, all other metals are reactive and react with the environment and form stable compounds, as their oxides, sulphides, chlorides and carbonates. They exist in their form of stable compounds called ores and minerals.

Examples: Iron Oxide (Fe_2O_3), Zinc Oxide (ZnO), Lead Sulfide (PbS), Calcium Carbonate ($CaCO_3$).

In their elemental state, metals have an oxidation number of 0. When they react with the environment, they oxidize, increasing their oxidation number. For example, iron (Fe) forms iron oxide (Fe_2O_3), with iron having an oxidation number of +3. This electron loss leads to the formation of stable compounds, called ores and minerals, as metals are more stable in these forms than as free elements.

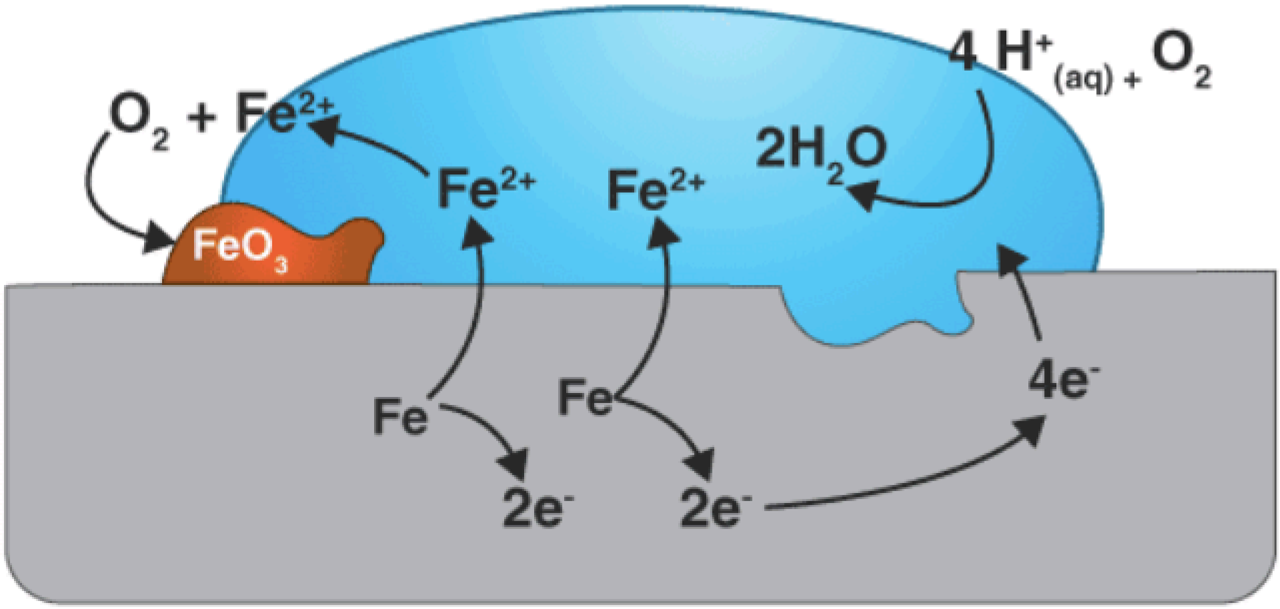

Figure 2. The electron flow mechanism involved in rust formation.

1.2.4. How and Why Corrosion Occurs?

- The metals are extracted from these compounds (ores). During the extraction, these ores are reduced to their metallic states.
- In the pure metallic state, the metals are unstable as they are considered in excited state ie., higher energy state.
- Therefore, as soon as the metals are extracted from their ores, the reverse process begins and forms metal compounds, which are thermodynamically stable, ie., lower energy state.
- When metals are used in various forms, they are exposed to the environment, (such as dry gases, moisture, etc.,) the exposed metal surface begins to decay ie., conversion into a more stable compound. This is the basic reason for metallic corrosion.

1.2.5. Effects of Corrosion:

Physical Property Degradation: Corrosion reduces electrical conductivity, malleability, and mechanical strength of metals.

- Electrical Conductivity: Corrosion forms a layer of oxides or other compounds on the metal surface, which disrupts electron flow and diminishes conductivity.
  For instance, when copper corrodes, it produces copper oxide, which conducts electricity poorly compared to pure copper.
- Malleability: Corrosion often makes metals more brittle, reducing their ability to deform without breaking.
  In the case of iron, rusting forms iron oxides that are flaky and fragile, restricting the metal's capacity to be shaped or bent.

- Mechanical Strength: As corrosion advances, the metal weakens, becoming more prone to failure under stress.
  For example, when steel corrodes due to exposure to water and air, its tensile strength decreases, making it more susceptible to cracking or breaking when under load.

1.2.6. Corrosion Products:

Corrosion Product refers to the substances formed when metals undergo corrosion, typically through chemical reactions with environmental factors such as oxygen, moisture, pollutants, or saltwater. These products are often compounds like oxides, sulphides, or carbonates that result from the metal's interaction with its surroundings.

Rusting of Iron: $Fe + O_2 + H_2O \rightarrow Fe_2O_3 \cdot nH_2O$
Tarnishing of Silver: $Ag + H_2S \rightarrow Ag_2S$
Green Layer on Copper: $Cu + CO_2 + H_2O \rightarrow CuCO_3 \cdot Cu(OH)_2$

1. **Rust (Iron Oxide):** When iron reacts with oxygen and water, it forms iron oxide (rust), a common corrosion product.
2. **Tarnishing of Silver:** Silver tarnishes when it reacts with hydrogen sulfide in the air to form silver sulfide. The tarnish appears as a black or dark grey layer on silver objects.
3. **Green Layer on Copper (Copper Carbonate):** Copper reacts with carbon dioxide and water in the air to form copper carbonate, creating a greenish layer on its surface.This greenish layer is often seen on copper roofs or statues, such as the Statue of Liberty.

These corrosion products can degrade the metal's properties, such as electrical conductivity, mechanical strength, and appearance.

1.2.7. Consequence or effects of corrosion:

1. **Efficiency Loss:** Corrosion products form over machinery, causing the equipment to lose efficiency.
2. **Contamination:** The products of corrosion can contaminate equipment or materials.
3. **Frequent Replacements:** Corroded equipment must be replaced frequently, leading to increased costs.
4. **Plant Failure:** Corrosion can lead to failures in industrial plants.
5. **Design Adjustments:** It may be necessary to over-design machinery and structures to compensate for the effects of corrosion.
6. **Health Hazards:** Corrosion releases toxic products, posing health risks.
7. **Delays:** Time is lost in acquiring replacement parts for corroded equipment.
8. **Value Reduction:** The value of goods is reduced due to corrosion.

DISCUSSION QUESTIONS

- Compare the energy and stability of pure metals and their metallic compounds
- Certain metals are affected by atmospheric gases while others are not. Reason out
- How does the corrosion of electrical wire affect its electrical conductivity?
- Why are metals like gold and silver, found in their pure form in nature, less prone to corrosion compared to metals like iron, which are often used in tools and machinery?